

SURAT, India

WMO: 428400

Lat: 21.200N

Lon: 72.833E

Elev: 39

StdP: 14.68

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	57.5	59.7	37.9	33.5	77.6	41.1	38.0	76.6	10.7	75.3	9.0	76.1	1.3	0	0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	11.4	100.7	72.6	98.1	73.2	96.0	73.9	83.1	89.7	82.6	89.0	81.9	88.2	3.9	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.5	164.1	86.6	80.9	160.6	86.1	80.3	157.7	85.7	47.2	89.8	46.6	89.3	46.0	88.5	85.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	9.8	8.0	53.4	105.5	2.9	2.4	51.3	107.2	49.5	108.7	47.9	110.1	45.7	111.8		
(5)			48.3	84.4	2.8	0.8	46.3	85.0	44.6	85.5	43.1	86.0	41.0	86.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	73.0	76.7	83.0	86.8	88.4	86.7	83.4	82.7	83.5	84.4	80.5	75.3	(6)
(7)		DBStd	5.48	3.44	4.26	3.76	2.83	2.30	3.07	2.36	1.85	2.22	2.69	3.01	3.38	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	1	0	1	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11699	714	748	1022	1104	1190	1101	1035	1014	1005	1066	916	784	(10)
(11)		CDD65	6225	249	329	557	654	725	651	570	549	555	601	466	319	(11)
(12)		CDH74	73139	1912	2978	6435	8682	10142	8891	6842	6234	6513	7310	4688	2512	(12)
(13)		CDH80	33517	584	1279	3332	4649	5690	4624	2491	1928	2383	3550	2112	895	(13)
(14)	Wind	WSAvg	2.9	2.4	2.2	2.2	2.9	4.8	5.0	4.5	3.9	2.3	1.4	1.6	2.0	(14)
(15)	Precipitation	PrecAvg	49.3	0.0	0.0	0.0	0.0	0.1	10.2	17.6	11.7	6.9	0.8	0.7	0.1	(15)
(16)		PrecMax	90.5	0.0	0.4	1.4	0.2	1.0	33.0	34.8	32.7	15.7	5.8	6.9	1.5	(16)
(17)		PrecMin	22.2	0.0	0.0	0.0	0.0	0.0	0.0	1.9	1.5	0.6	0.0	0.0	0.0	(17)
(18)		PrecStd	16.7	0.0	0.1	0.3	0.0	0.3	7.9	8.3	8.0	4.4	1.5	1.5	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.2	96.7	102.4	104.3	103.7	97.0	91.9	89.9	94.5	98.1	95.3	91.7	(19)
(20)			MCWB	67.9	68.6	70.4	69.9	75.0	80.9	80.9	79.9	79.9	75.3	72.0	69.1	(20)
(21)		2%	DB	87.9	93.3	99.0	101.2	99.2	94.4	89.6	88.2	91.7	96.1	93.2	89.5	(21)
(22)			MCWB	66.6	68.4	69.2	72.0	76.6	81.1	81.0	79.7	79.6	75.0	71.2	68.6	(22)
(23)		5%	DB	85.6	90.6	96.3	98.6	95.8	93.0	88.2	87.0	89.8	94.4	91.6	87.4	(23)
(24)			MCWB	66.1	67.8	69.3	72.9	78.4	81.2	80.8	79.5	79.5	74.7	71.2	68.1	(24)
(25)		10%	DB	83.3	87.8	93.8	95.8	93.2	91.7	87.0	86.0	88.1	92.6	89.9	85.3	(25)
(26)			MCWB	65.2	66.8	69.1	73.0	79.8	81.1	80.6	79.3	79.0	75.0	71.0	67.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.7	74.0	78.5	81.6	83.7	83.8	83.0	82.2	82.8	82.9	79.9	74.7	(27)
(28)			MCDWB	84.1	86.5	87.3	90.7	90.3	90.8	88.3	86.5	88.8	90.1	88.5	86.0	(28)
(29)		2%	WB	69.8	72.1	76.5	80.5	82.8	83.3	82.1	81.0	81.8	81.7	78.3	72.5	(29)
(30)			MCDWB	82.6	84.1	85.2	88.5	89.7	90.1	86.9	85.2	87.3	88.5	85.9	84.0	(30)
(31)		5%	WB	68.5	70.9	74.9	79.6	82.2	82.7	81.6	80.6	81.1	80.8	76.6	70.7	(31)
(32)			MCDWB	80.9	83.1	83.9	87.5	89.3	89.4	86.2	84.7	86.2	87.5	84.2	82.4	(32)
(33)		10%	WB	67.1	69.4	73.6	78.8	81.6	82.1	81.0	80.0	80.5	79.9	74.7	69.1	(33)
(34)			MCDWB	79.5	81.8	83.3	86.5	88.8	88.7	85.4	84.1	85.4	86.5	83.5	81.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.6	22.3	21.7	17.9	11.4	8.3	6.0	6.1	9.0	17.4	20.7	22.0	(35)
(36)			MCDBR	25.1	25.9	25.5	23.4	16.4	9.9	7.7	7.6	11.7	21.5	23.0	24.2	(36)
(37)			MCWBR	9.8	9.0	8.2	7.6	5.8	3.3	3.3	3.0	4.2	7.1	8.1	9.4	(37)
(38)		5% WB	MCDBR	21.8	21.8	20.2	15.7	11.3	9.0	7.1	6.7	10.2	13.7	16.6	21.5	(38)
(39)			MCWBR	10.1	9.1	7.5	6.1	3.8	3.6	3.6	3.2	4.6	6.0	7.5	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.449	0.443	0.491	0.537	0.572	0.686	0.701	0.644	0.595	0.565	0.539	0.495	(40)	
(41)		taud	2.116	2.148	2.029	1.961	1.927	1.717	1.695	1.813	1.911	1.920	1.934	1.989	(41)	
(42)		Ebn at Noon	258	269	260	250	239	211	208	221	230	230	228	237	(42)	
(43)		Edh at Noon	46	47	55	60	62	75	77	69	62	59	55	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1512	1791	2091	2285	2307	1742	1239	1267	1529	1718	1517	1418	(44)	
(45)		RadStd	46	51	56	40	72	144	140	117	133	85	71	66	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page